Newton's Method - a first view

$\min: f(x)$

Gradient descent: linear approximation

$$f(x+d) \approx \underbrace{f(x) + \nabla f(x)^T (d)}_{\text{linear function}} + \frac{1}{2\eta}\|d\|_2^2$$

Convergence rate:

If f is β-smooth, α-strongly convex

then GD : $O\left(\frac{\beta}{\alpha} \cdot \log(1/\varepsilon)\right)$ – linear convergence

Newton's Method - a first view

$f: R^n \rightarrow R$ $\qquad$ $\left(\nabla^2 f(x)\right)_{i,j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$

Newton: quadratic approximation

$$f(x+d) \simeq f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x) d$$

Convergence rate:

$O\left(\log \log (1/\varepsilon)\right)$ $\qquad$ quadratic convergence

Newton vs Gradient Descent: computational comparison

$$f(x+d) \approx \underbrace{f(x) + \nabla f(x)^T d + \tfrac{1}{2} d^T \nabla^2 f(x)}_{\text{convex quadratic function}} d$$

Take deriv w.r.t. d, set $=0$, solve for d

$$\nabla f(x) + \nabla^2 f(x) \cdot d = 0$$

$$\Rightarrow d = -\left(\nabla^2 f(x)\right)^{-1} \nabla f(x)$$

Newton: $x_{t+1} = x_t + d = x_t - \left(\nabla^2 f(x_t)\right)^{-1} \nabla f(x_t)$

Grad. Desc: $x_{t+1} = x_t - \eta \nabla f(x_t)$.

GD: vector addition : $O(n)$

Newton: matrix inversion : $O(n^3)$

GD: $O(n \cdot (\beta/\alpha) \cdot \log(1/\epsilon))$ vs Newton: $O(n^3 \cdot \log\log(1/\epsilon))$

The Newton Direction

$$f(x+d) \approx f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x) d$$

$$d_N = -\left(\nabla^2 f(x)\right)^{-1} \nabla f(x)$$

Newton's: $x_{t+1} = x_t - \left(\nabla^2 f(x)\right)^{-1} \nabla f(x)$

no step size ?

Algorithm: damped and undamped

Damped: $x_{t+1} = x_t - \eta \left(\nabla^2 f(x_t)\right)^{-1} \nabla f(x_t)$

Undamped: $x_{t+1} = x_t - \left(\nabla^2 f(x_t)\right)^{-1} \nabla f(x_t)$

Newton's method: 2 phases

Phase 1: When not v. close to x^*, use damped Newton

Phase 2: Close to x^*, undamped // quadratic convergence.